Where Energy Recovery Stalls

Two bottlenecks in Japan's municipal waste-incineration fleet

15-18 MINUTE ROUTE | DATA + METHOD | RESULTS + LIMITS

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Paper discussion briefing

Why This Matters

1

Waste is burned

Japan relies heavily on municipal waste incineration.

→

2

Heat already exists

Incineration creates heat whether or not electricity is recovered.

→

3

Not all heat becomes electricity

In FY2024, only 41.1% of panel facilities are flagged as power-generating.

> The same waste-treatment process can either recover useful power or miss that energy-recovery opportunity.

The Real Question

EXPECTED PART

Younger and larger plants have advantages. That is not the surprising claim.

The real question is whether energy-recovery modernization spreads through lagging facilities, or concentrates where conditions are already favorable.

Two Questions, One Fleet

CORE LOGIC

One fleet average hides two different bottlenecks.

QUESTION 1

Which plants start generating electricity?

This is the entry bottleneck.

QUESTION 2

Among generators, who produces more electricity per tonne?

This is the performance bottleneck.

What the Data Can See

Observed layer	Observed fields	Analytical role
MOE General Waste Treatment Survey	National municipal waste-treatment facility records, FY2005-FY2024	Base facility-year panel and power-generation outcome
Facility identifiers and location	Official code, facility name, prefecture, fiscal year	Panel linkage, prefecture controls, clustered uncertainty, duplicate-code checks
Facility operating attributes	Power-generation flag/output, throughput, design capacity, start year/age	Entry risk set and generator-output comparison
Context and plausibility controls	Heating value and grid-emission context	Comparability checks, not the central contribution

> The panel observes facility reporting patterns well. It does not directly observe internal retrofit contracts, municipal bargaining, or full lifecycle emissions.

How the Panel Becomes Two Frames

STARTING PANEL

23,599

facility-year rows

Full analytical starting point.

→

CODED FLEET

2,948

identifiable facilities

Official identifiers support panel linkage.

→

ENTRY FRAME

13,770

at-risk facility-years

2,035 initial non-generators; 141 first-entry events.

→

PERFORMANCE FRAME

5,683

generator observations

1,016 operating generators with covariates.

> The split separates two outcomes: crossing into power generation, and performing well after crossing.

Why Two Samples Are Needed

Question	Sample frame	Outcome	What it tests
Who starts generating?	Facilities first observed without power generation	First report of electricity generation	Whether energy recovery diffuses into the lagging fleet
Who generates well?	Identifiable operating generators	MWh generated per tonne processed	Whether performance converges after entry

> One model for everything would mix the gate into generation with performance after entry, hiding where the bottleneck actually sits.

Method: First Entry Into Generation

PLAIN MODEL

Pr(first generation in year t) = f(prior age, prior capacity, year, prefecture)

RISK SET

Only initial non-generators are candidates.

Facilities already generating in their first observed year are left-censored for first-entry analysis.

TIMING DISCIPLINE

Predictors are lagged.

Age and capacity are measured before first entry, not after the event.

COMPARABILITY

Year and prefecture controls.

Uncertainty is clustered by facility, and alternative hazard forms are checked.

Plain question: among facilities still outside power generation, who first reports generation in the next observed year?

Method: Output Among Generators

OUTCOME MODEL

log(MWh per tonne) = age + capacity + utilization + heating value + grid context + year structure

OUTCOME

Electricity recovered per tonne processed.

This measures output intensity, not just the existence of a generator.

MODEL FAMILY

Pooled OLS, year FE, RE, year FE + RE.

Multiple structures check whether the sign pattern survives.

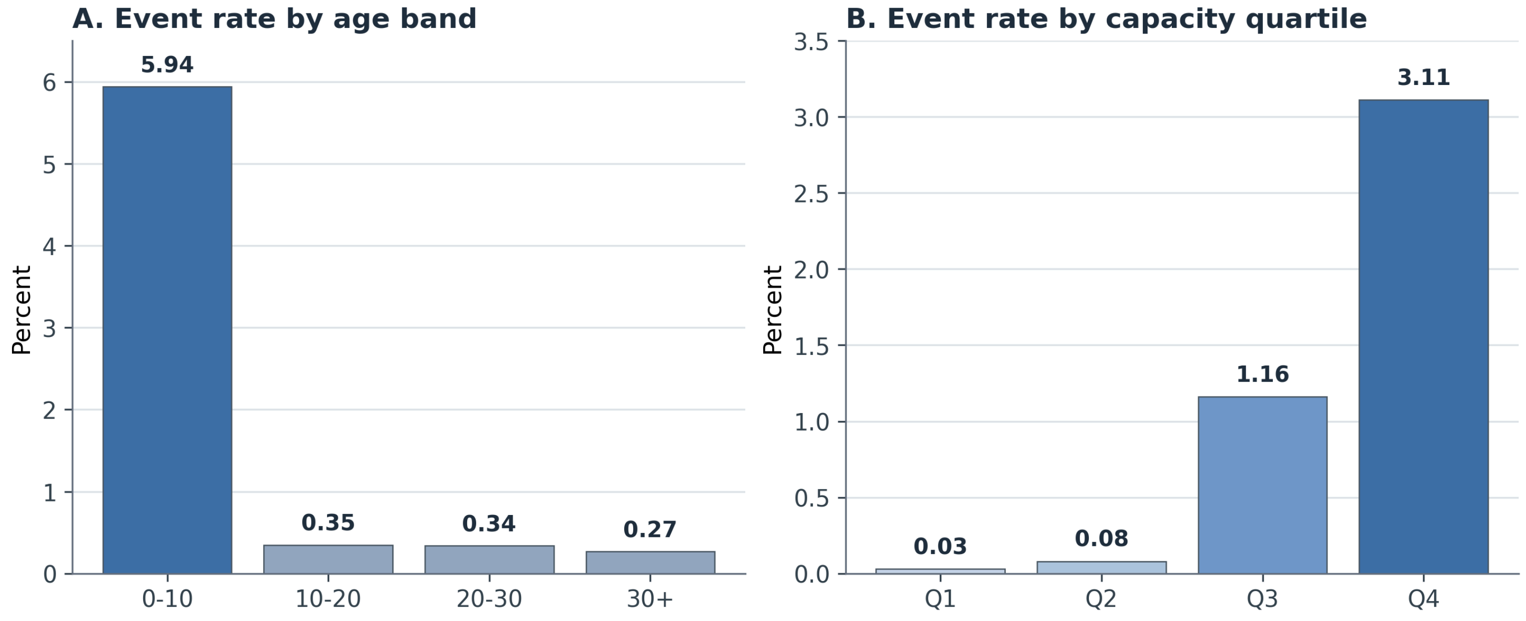
INTERPRETATION

This is diagnostic, not causal.

It tests whether generators converge enough to erase inherited facility differences.

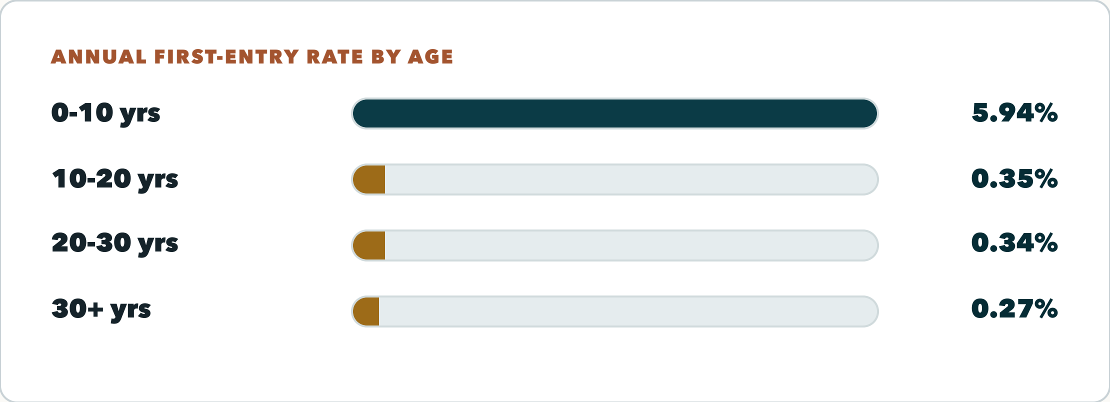
The critical question after entry is whether generator performance converges enough to erase inherited facility differences.

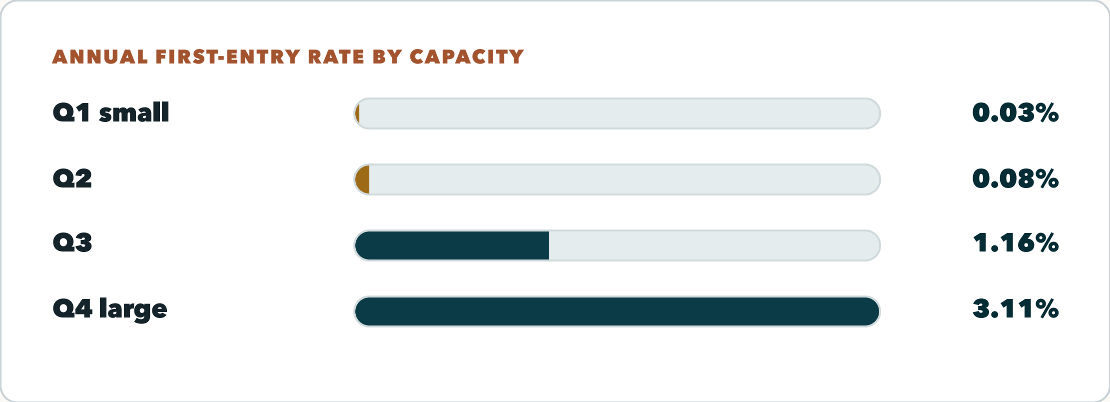
Result 1: Entry Clusters in Young, Large Facilities



First entry is concentrated in already favorable facility types: 102/141 events are age 0-10, and 99/141 are in the largest capacity quartile.

Interpretation: Not Broad Fleet Catch-Up





If modernization were broad catch-up, older and smaller non-generators would show more entry. Instead, the smallest capacity quartile records 1 event; the largest records 99.

Entry Pathways: Modernization, Not One Mechanism

RESET / REBUILD-LIKE

82

observed events

Capital-side modernization is empirically present.

CONTINUITY / UPGRADE-LIKE

38

observed events

Some cases remain consistent with in-place upgrade.

PLACEHOLDER / FORWARD-DATED

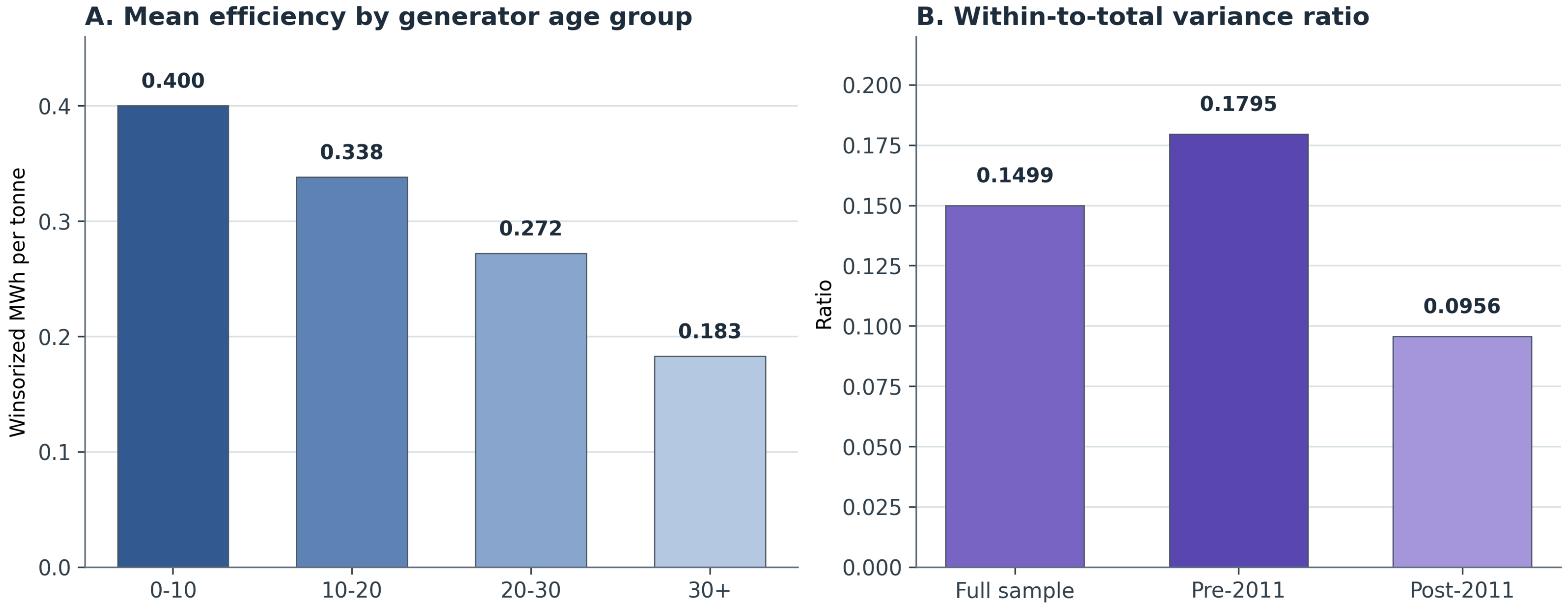
20

observed events

Some entries should not be forced into a stronger mechanism claim.

> The evidence supports selective modernization, but it does not prove that replacement is the only pathway.

Result 2: Generation Status Is Not Enough



Among generators, output intensity still differs sharply: age 0-10 averages 0.400 MWh/t, while age 30+ averages 0.183 MWh/t.

Interpretation: Performance Gaps Persist

FULL SAMPLE

0.1499

within-to-total variance ratio

Most generator-output variation is between facilities.

FY2005-FY2011

0.1795

within-to-total ratio

Baseline period for the Fukushima-context robustness split.

FY2012-FY2024

0.0956

within-to-total ratio

Post-Fukushima-context period; cross-facility hierarchy remains.

> The 2011 split is a context check around Fukushima, not a causal claim. In both periods, generation entry does not erase inherited facility gaps.

Stress Tests: The Pattern Survives

Stress test	Why it matters	What survives
Composite facility-ID sensitivity	Checks whether duplicate official codes drive the adoption result	Age penalties and positive capacity effect remain stable
Alternative adoption models	Checks whether logit hazard choice creates the result	Complementary log-log and LPM keep the same sign pattern
Generator-output stress tests	Checks period, scale, and outcome-coding sensitivity	Age stays negative; capacity and utilization stay positive
Heating-value plausibility restrictions	Checks whether noisy heat-value data drive model patterns	Main age, scale, and utilization coefficients remain stable

> These checks do not make the estimates causal. They show the diagnostic pattern is not a fragile artifact of one coding choice.

Data Limits: Disclosed and Tested

DUPLICATE-CODE CONCERN

39

official codes affected

444 source rows use codes that duplicate within at least one fiscal year.

MISSING-CODE CONCERN

907

operating-generator rows

Rows missing official codes are excluded from the canonical regression frame.

PLAUSIBILITY CONCERN

569

heating-value rows

Rows outside 3-25 MJ/kg are checked through sensitivity restrictions.

> These limits are disclosed and stress-tested. The evidence maps bottlenecks; it does not claim a perfect engineering census or fully identified causal mechanism.

Why the Split Matters for Decisions

A

Non-generators

Can this plant start recovering electricity?

→

B

Generators

Can this plant recover more electricity from the same waste?

→

C

Planning

Use different decisions for different bottlenecks.

Renewal, starting electricity generation, and generator optimization are different decisions, not one generic "improve incineration" task.

Contribution: Two Bottlenecks in One Fleet

WEAK VERSION

Newer and larger plants generate more.

This is plausible but not enough for a paper by itself.

STRONGER VERSION

The fleet has two bottlenecks.

Starting generation is selective, and generator output remains uneven after entry.

> The contribution is locating the bottlenecks, not pretending that age and scale advantages are surprising.

Next Step: Test Mechanisms

CAPITAL RENEWAL

Link entry to investment and rebuild histories.

This would test whether selective entry is mainly capital replacement or retrofit.

GOVERNANCE AND ROUTING

Link facilities to municipal decisions.

This could explain why some plants start generation or improve output.

CLIMATE AND COMPARISON

Add lifecycle or cross-fleet evidence.

This would move from energy-recovery diagnosis toward climate or international comparison.

> The next research step is moving from diagnostic mapping to mechanism testing.

Discussion Questions

Two bottlenecks shape the story: who starts generating electricity, and who generates well after starting.

Is the two-bottleneck explanation strong enough as the central contribution? Which limitation or future direction should be prioritized?